

Tensor Network Theory:

A formalization blueprint

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Chapter 1

Source preparations and local contractions

1.1 Elimination of normalized pair effects

This section treats [1, Lemma 5.1], September 24, 2026, 04-compression.tex, lines 53–127, together with the choice of the stack length and the rescaling in the proof of [1, Theorem 5.2], lines 201–212. The estimates are first proved for monomials written as chains of recorded pair effects between arbitrary contractions. The statements about parties are then proved after assigning every register to a party. Preparing the sources before the local operations and combining them by unordered pairs gives the full conclusion of the source lemma.

Lemma 1.1.1 (Elimination of normalized pair effects). Lean ✓ Stmt *Let G be a contraction with an expansion $G = \sum_{\xi} c_{\xi} M_{\xi}$ into K allowed monomials on a fixed set of parties, each with at most r pair effects. For every integer $m \geq 1$ there are a gate $\tilde{G}_m$ with additional garbage registers and a fixed normalized garbage vector Γ_m with $\|\tilde{G}_m - G \otimes |\Gamma_m\rangle\| \leq (r/\sqrt{m}) \sum_{\xi} |c_{\xi}|$. The gate $\tilde{G}_m$ has an expansion using only local contractions and normalized pair sources, with at most Km^r monomials and absolute coefficient sum at most $\sum_{\xi} |c_{\xi}|$. Its additional registers are owned by the original participating parties, and all sources on the same pair of parties may be combined into one normalized pair source. This is [1, Lemma 5.1].*

Proof. ✓ Proof The error, normalization, expansion, and ownership conclusions follow from Theorem 1.1.2. Apply Theorem 1.2.5 to each monomial in this expansion. It replaces all sources on a given unordered pair by one normalized source and recovers the original registers by local isometries and exchanges. The operator of each monomial is unchanged, as are the number of monomials, their coefficients, and their common output space. $\square$

Theorem 1.1.2 (Elimination of pair effects on parties). Lean ✓ Stmt *Let $G = \sum_{\xi} c_{\xi} M_{\xi}$ be an expansion into K allowed monomials on parties, each with at most r pair effects, and let $m \geq 1$. Then Γ_m is a unit vector prepared by normalized pair sources, and $\|\tilde{G}_m - G \otimes |\Gamma_m\rangle\| \leq (r/\sqrt{m}) \sum_{\xi} |c_{\xi}|$. Read on the registers of the parties, $\tilde{G}_m$ is a weighted sum of at most Km^r operators of allowed source-only compositions, with absolute coefficient sum $\sum_{\xi} |c_{\xi}|$. Its additional registers are the stacks of the effect occurrences, and the stack of an effect on the parties (p, q) consists of m registers of p and m registers of q . This is [1, Lemma 5.1] without its last clause,*

that all sources on one pair of parties may be combined into one, and without the hypothesis that G is a contraction, which is not needed. The participating parties are the endpoint parties of the effects of the monomials.

Proof. ✓ Proof The error bound, the count and the coefficient sum are Theorem 1.1.11, applied to the monomials placed on the parties. The expansion consists of compositions of local contractions and normalized pair sources by Theorem 1.1.16, and the additional registers are the stacks of Theorem 1.1.17. □

Throughout, η is a unit vector of a finite-dimensional Hilbert space $\mathcal{W}$, typically a pair space $\mathcal{W} = \mathcal{U} \otimes \mathcal{V}$, and $m \geq 1$.

Definition 1.1.3 (Cyclic insertion map). Lean Lean Lean ✓ Stmt For a position $k \in \{1, \dots, m\}$, let $I_k : \mathcal{W} \rightarrow \mathcal{W}^{\otimes m}$ place its input in register k and a copy of η in every other register. The cyclic insertion map is the average

$$T_m = \frac{1}{m} \sum_{k=1}^m I_k,$$

and the ideal insertion map is $P_m = |\eta^{\otimes m}\rangle\langle\eta|$.

Theorem 1.1.4 (Insertions as register permutations). Lean Lean Lean ✓ Stmt For a permutation σ of the registers, $I_{\sigma(k)}$ is I_k followed by the corresponding permutation of the tensor factors of $\mathcal{W}^{\otimes m}$. Hence T_m is the average of the m cyclic register permutations applied after the insertion in the first register. For a pair space $\mathcal{W} = \mathcal{U} \otimes \mathcal{V}$, the index map of a register permutation permutes the $\mathcal{U}$ indices and the $\mathcal{V}$ indices by the same permutation.

Proof. ✓ Proof Both sides agree on product vectors, coordinate by coordinate. The pair statement is the corresponding identity of index maps. □

Theorem 1.1.5 (Contractivity and the fixed vector). Lean Lean ✓ Stmt The map T_m is a contraction, and $T_m \eta = \eta^{\otimes m}$.

Proof. ✓ Proof Each I_k is an isometry because $\|\eta\| = 1$, so the average of the m maps has norm at most one. Each I_k sends η to $\eta^{\otimes m}$, and the average of m equal terms is that term. □

Theorem 1.1.6 (One-effect estimate). Lean Lean Lean Lean Lean ✓ Stmt For every $v \perp \eta$ one has $\|T_m v\| = \|v\|/\sqrt{m}$. Consequently $\|T_m - P_m\| \leq m^{-1/2}$, with equality when the orthogonal complement of η is nonzero, and $T_m = P_m$ when it is zero. The same upper bound holds for $(T_m - P_m) \otimes \mathbf{1}_{\mathcal{S}}$ on any spectator space $\mathcal{S}$.

Proof. ✓ Proof For $v \perp \eta$, the vectors $I_j v$ and $I_k v$ with $j \neq k$ are orthogonal, since their inner product contains the factor $\langle v, \eta \rangle = 0$, and each has norm $\|v\|$. Hence $\|\sum_k I_k v\|^2 = m\|v\|^2$ and $\|T_m v\| = \|v\|/\sqrt{m}$. For a general input v , the difference $(T_m - P_m)v$ equals T_m applied to the component $v - \langle \eta, v \rangle \eta$ orthogonal to η , whose norm is at most $\|v\|$. This gives the upper bound. A nonzero vector orthogonal to η attains it, and when no such vector exists the difference vanishes. Tensoring with the identity does not increase the operator norm. □

Definition 1.1.7 (Monomials with recorded pair effects). Lean Lean Lean Lean Lean ✓ Stmt A monomial with n recorded pair effects is a composition

$$M = F_n \circ E_n \circ F_{n-1} \circ \dots \circ E_1 \circ F_0,$$

where each F_i ends with a pair register $\mathcal{W}_i$ beside a spectator space $\mathcal{S}_i$, and $E_i = \langle \eta_i | \otimes \mathbf{1}_{\mathcal{S}_i}$ contracts that register with a pair bra. The monomial is allowed when every F_i is a contraction and every η_i is a unit vector. Here the pieces F_i are arbitrary contractions; no division of registers among parties is recorded.

Theorem 1.1.12 (Combining pair sources). [Lean](#) [Lean](#) [✓ Stmt](#) For pair vectors $\eta \in \mathcal{U} \otimes \mathcal{V}$ and $\eta' \in \mathcal{U}' \otimes \mathcal{V}'$, regard $\eta \otimes \eta'$ as a pair vector in $(\mathcal{U} \otimes \mathcal{U}') \otimes (\mathcal{V} \otimes \mathcal{V}')$. Its norm is $\|\eta\| \|\eta'\|$; in particular, normalized pair vectors combine into a normalized pair vector.

Proof. [✓ Proof](#) The squared norm is a sum of products of squared coordinates, which factorizes after regrouping the index set. $\square$

Definition 1.1.13 (Registers of parties and source-only compositions). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) A register is a Hilbert space together with the party that owns it. The memory of a finite list of registers is the tensor product of their spaces. A source-only composition is a composition of the following operations: a local map from some registers of one party p to new registers of p , tensored with the identity on all other registers; the preparation of a vector $\eta \in \mathcal{U} \otimes \mathcal{V}$ on two new registers $\mathcal{U}$ of a party p and $\mathcal{V}$ of a party $q \neq p$; and reorderings of the tensor factors. It is allowed when its local maps are contractions and its prepared pair vectors are unit vectors. An operator between two memories is source-only when it is the operator of an allowed source-only composition.

Definition 1.1.14 (Allowed monomials on parties). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) An allowed monomial on parties is a chain of source-only compositions separated by pair effects. Each pair effect contracts a register $\mathbb{C}^\alpha$ of a party p and a register $\mathbb{C}^\beta$ of a party $q \neq p$ with a bra $\langle \eta |$, $\eta \in \mathbb{C}^\alpha \otimes \mathbb{C}^\beta$; the monomial is allowed when its source-only compositions are allowed and every η is a unit vector. An effect on arbitrary finite-dimensional registers of p and q takes this form after local isometric identifications with $\mathbb{C}^\alpha$ and $\mathbb{C}^\beta$ and a reordering of tensor factors, absorbed into the preceding source-only composition. As an operator it is a monomial with recorded pair effects in the sense of Definition 1.1.7, whose pieces between effects are source-only. This is the notion of allowed monomial of [1, Section 5.1], lines 32–35.

Theorem 1.1.15 (Insertion by pair sources). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $\eta \in \mathbb{C}^\alpha \otimes \mathbb{C}^\beta$ be a unit vector on the parties $p \neq q$, and consider a stack of m pair registers, each made of a register $\mathbb{C}^\alpha$ of p and a register $\mathbb{C}^\beta$ of q . Preparing pair sources of η in the positions other than k , with the input pair register in position k , is a source-only composition, and it equals the insertion map I_k . Preparing η in all m positions gives $\eta^{\otimes m}$.

Proof. [✓ Proof](#) By induction on m : for k the first position, the remaining $m - 1$ positions are prepared; otherwise a source of η is prepared in the first position and the insertion at position $k - 1$ is applied to the remaining registers. $\square$

Theorem 1.1.16 (The expansion is source-only). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $G = \sum_{\xi} c_{\xi} M_{\xi}$ be an expansion into allowed monomials on parties. Identify the stack of each effect occurrence on (p, q) with m pair registers whose $\mathbb{C}^\alpha$ halves belong to p and whose $\mathbb{C}^\beta$ halves belong to q . Then every term of the expansion of $\tilde{G}_m$ is source-only, and the garbage vector Γ_m is prepared by normalized pair sources on the endpoint parties of the effects.

Proof. [✓ Proof](#) In a term, each effect is replaced by an insertion, which is source-only by Theorem 1.1.15, and the remaining pieces are the source-only compositions of the monomial, now acting next to the idle stacks. The stacks of the other monomials are prepared in their ideal content $\eta^{\otimes m}$, again by pair sources. Both statements follow by induction on the monomial and on the list of monomials. $\square$

Theorem 1.1.17 (Ownership of the additional registers). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The output registers of $\tilde{G}_m$ are the output registers of G together with the stacks of all effect occurrences of all monomials. The stack of an effect on the parties (p, q) consists of m pair registers, each made of one register of p and one register of q .

Proof. ✓ Proof By induction on the monomials, collecting the stacks. □

Theorem 1.1.18 (Combining pair sources on one pair of parties). Lean Lean Lean Lean
✓ Stmt Let $\eta \in \mathcal{U} \otimes \mathcal{V}$ and $\eta' \in \mathcal{U}' \otimes \mathcal{V}'$ be pair vectors on the parties $p \neq q$. Preparing η' and then η , and merging the two new registers of p into $\mathcal{U} \otimes \mathcal{U}'$ and the two new registers of q into $\mathcal{V} \otimes \mathcal{V}'$, is the preparation of the regrouped vector $\eta \otimes \eta' \in (\mathcal{U} \otimes \mathcal{U}') \otimes (\mathcal{V} \otimes \mathcal{V}')$, whose norm is $\|\eta\| \|\eta'\|$. For coordinate spaces the norm identity is Theorem 1.1.12.

Proof. ✓ Proof Both operators agree on product vectors, and the regrouping is an isometry. □

Corollary 1.1.19 (Stack length and rescaling). Lean Lean Lean Lean ✓ Stmt In the setting of Theorem 1.1.11, suppose moreover that G is a contraction, and let $\delta > 0$ and $S \geq \sum_{\xi} |c_{\xi}|$.

With $m = \lceil (rS/\delta)^2 \rceil + 1$, the replaced gate satisfies $\|\tilde{G}_m - G \otimes |\Gamma_m\rangle\| \leq \delta$ and $\|\tilde{G}_m\| \leq 1 + \delta$, and $\tilde{G}_m/(1 + \delta)$ is a contraction within 2δ of $G \otimes |\Gamma_m\rangle$. The number m depends only on r , S and δ .

Proof. ✓ Proof The choice of m gives $\sqrt{m} \geq rS/\delta$, so the bound of Theorem 1.1.11 is at most δ . Since $G \otimes |\Gamma_m\rangle$ is a contraction, the triangle inequality gives $\|\tilde{G}_m\| \leq 1 + \delta$. Dividing by $1 + \delta$ yields a contraction, and its distance to $G \otimes |\Gamma_m\rangle$ is at most $\delta/(1 + \delta) + \delta \leq 2\delta$. □

1.2 Preparing and combining pair sources

Theorem 1.2.1 (Exchanging lists of registers). Lean Lean Lean Lean Lean Lean Lean
Lean Lean ✓ Stmt Let $\mathcal{H}_A$, $\mathcal{H}_B$, and $\mathcal{H}_C$ be the memories of three ordered lists of registers. Adjacent exchanges give an allowed composition that sends $a \otimes (b \otimes c)$ to $b \otimes (a \otimes c)$. The order within each list is preserved. This composition contains no pair-source preparations. Taking the second list to consist of one register gives the corresponding move of that register to the front.

Proof. ✓ Proof Move the first register of the second list past the first list by adjacent exchanges, then repeat for the remaining registers of the second list. Induction on the lists gives the tensor identity. □

Definition 1.2.2 (The sources of a composition). Lean Lean Lean Lean Lean Lean Lean
Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt A pair source consists of distinct parties p, q , their Hilbert spaces $\mathcal{U}, \mathcal{V}$, and a vector $\eta \in \mathcal{U} \otimes \mathcal{V}$. Its two registers remain assigned to p and q , respectively. For a finite list $S = (s_1, \dots, s_m)$ of pair sources, let $\mathcal{H}_S$ be the tensor product of their registers in list order. Its preparation on a memory $\mathcal{H}$ is the map

$$P_{S, \mathcal{H}} : \mathcal{H} \longrightarrow \mathcal{H}_S \otimes \mathcal{H}, \quad x \longmapsto (\eta_1 \otimes \dots \otimes \eta_m) \otimes x. \quad (1.1)$$

Tensor products are identified by the prescribed associativity maps. The source list of a composition records each occurrence separately, including repetitions of the same vector on the same pair of parties. This is the source data used in [1, proof of Theorem 5.2], `04-compression.tex`, lines 233–251.

Theorem 1.2.3 (Preparation before the local operations). Lean Lean Lean Lean Lean
Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Let $F : \mathcal{H}_{\text{in}} \rightarrow \mathcal{H}_{\text{out}}$ be an allowed composition of local contractions, pair-source preparations, and changes of tensor-factor order. Let S be its source list. There is an allowed composition $B : \mathcal{H}_S \otimes \mathcal{H}_{\text{in}} \rightarrow \mathcal{H}_{\text{out}}$ consisting of local contractions and changes of tensor-factor order, with no source preparations, such that

$$F = B \circ P_{S, \mathcal{H}_{\text{in}}}. \quad (1.2)$$

Every vector in S is normalized. The source occurrences and their endpoint parties are those of the original composition.

Proof. ✓ Proof Induct on the composition. A local operation, an exchange, or the identity contributes no source. For a pair-source preparation, take that source as the entire list and take the remaining operation to be the identity. In the composition case, move a later preparation before an earlier operation $A : \mathcal{H} \rightarrow \mathcal{H}'$ by adjoining its untouched registers. The required identity is

$$P_{S, \mathcal{H}'} \circ A = (\text{id}_{\mathcal{H}_S} \otimes A) \circ P_{S, \mathcal{H}}.$$

It follows first on elementary tensors and then by linearity. When an untouched register precedes the composition, exchanges move it in front of the newly prepared source registers before the remaining local operations act. These exchanges are isometries. Tensoring each original local map with an identity preserves its contraction bound. $\square$

Definition 1.2.4 (Expansion of a preparation). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt A source list G expands to S when, beside every spectator memory $\mathcal{H}$, there is an allowed composition E with no source preparations such that $E \circ P_{G, \mathcal{H}} = P_{S, \mathcal{H}}$. Expansion is reflexive and transitive, and remains valid upon adjoining a common source. Exchanging two successive sources, or reversing the endpoint order of a source and exchanging its two half-spaces, gives such an expansion.

Theorem 1.2.5 (One source for each participating pair). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Let S be a finite list of normalized pair sources. There is a list G of normalized pair sources containing exactly one source for each unordered pair of parties occurring in S , and an allowed composition $E : \mathcal{H}_G \otimes \mathcal{H} \rightarrow \mathcal{H}_S \otimes \mathcal{H}$ with no source preparations, such that

$$P_{S, \mathcal{H}} = E \circ P_{G, \mathcal{H}}. \tag{1.3}$$

Consequently every allowed composition admits (1.2) with at most one normalized source on each participating pair.

Proof. ✓ Proof Insert the sources successively. Exchange tensor factors to bring together the sources with a prescribed unordered pair of owners; reverse the order of the two half-spaces when necessary. Replace two such vectors, on $U \otimes V$ and $U' \otimes V'$, by the image of their tensor product under the canonical isometry $(U \otimes V) \otimes (U' \otimes V') \rightarrow (U \otimes U') \otimes (V \otimes V')$. Its norm is the product of the two original norms and is therefore one. Splitting the two local tensor factors and exchanging the middle registers recovers the original preparations by isometries. Continuing through the finite list proves (1.3); no other pair of owners is introduced. Compose this expansion with the local operations in Theorem 1.2.3. $\square$

1.3 Collecting local operations by party

Definition 1.3.1 (The registers of a party). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt For an ordered list of registers ℓ and a party p , let ℓ_p retain precisely the registers owned by p , in their original order. Write $\mathcal{H}_{\ell_p}$ for their joint memory. The memory of an empty list is $\mathbb{C}$. A finite ordered list of distinct parties $p_1, \dots, p_s$ gives the grouped memory $\bigotimes_{j=1}^s \mathcal{H}_{\ell, p_j}$. Local maps between these memories act by their ordered tensor product; for an

empty party list this is the identity on $\mathbb{C}$. Equal register lists are identified by the identity on their memories, preserving operator norms.

The parties of a composition include every register owner and the party assigned to each local operation. In particular, a scalar operation $\mathbb{C} \rightarrow \mathbb{C}$ retains its assigned party when both its input and output register lists are empty. These are the local spaces and operations in [1, Theorem 5.2], 04-compression.tex, lines 233–251.

Definition 1.3.2 (Canonical partitions and grouping). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt For a choice of parties A , there is a prescribed isometric identification

$$U_{A,\ell} : \mathcal{H}_\ell \longrightarrow \mathcal{H}_{\ell|_A} \otimes \mathcal{H}_{\ell|_{A^c}}, \quad (1.4)$$

which retains the order within each selected list. It is obtained solely from associativity, exchanges of tensor factors, and the unit identifications with $\mathbb{C}$.

If $p_1, \dots, p_s$ are distinct and contain every owner in ℓ , successive partitions give a prescribed isometric identification $U_\ell : \mathcal{H}_\ell \rightarrow \bigotimes_{j=1}^s \mathcal{H}_{\ell, p_j}$.

Proof. ✓ Proof Induct on ℓ . Its first register belongs to exactly one of the two parts; associativity places it first in that part and an exchange moves it past the other part when necessary. For the second assertion, successively separate the registers of $p_1, \dots, p_s$. Distinctness prevents any register from being repeated, and coverage makes the final remaining register list empty. $\square$

Theorem 1.3.3 (Restriction to selected parties). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Let F be a composition containing no pair-source preparations. For any choice of parties A , retaining their registers and local operations gives a composition F_A containing no pair-source preparations. It is allowed whenever F is allowed. Its local operations occur in their original chronological order, and its parties are precisely those of F that belong to A .

Proof. ✓ Proof Retain a local operation exactly when its assigned party belongs to A . Restrict every exchange to the retained registers, and restrict compositions and untouched registers recursively. The local contraction bounds are unchanged. Induction on the composition proves the assertion about its parties, including scalar local operations. $\square$

Theorem 1.3.4 (Factorization across a partition of the parties). Lean ✓ Stmt If $F : \mathcal{H}_\ell \rightarrow \mathcal{H}_{\ell'}$ contains no pair-source preparations, then for any choice of parties A its restrictions satisfy

$$U_{A,\ell'} F = (F_A \otimes F_{A^c}) U_{A,\ell}. \quad (1.5)$$

This identity does not require bounds on the norms of the local maps.

Proof. ✓ Proof Induct on the composition. A local operation acts wholly in one part. An exchange either stays within one part or changes only the prescribed ordering between the parts. Tensoring with an untouched register and composition preserve the identity. $\square$

Theorem 1.3.5 (Tensor product of the local maps). Lean Lean Lean ✓ Stmt Let $F : \mathcal{H}_\ell \rightarrow \mathcal{H}_{\ell'}$ be an allowed composition containing no pair-source preparations. Let $p_1, \dots, p_s$ be distinct parties containing every party of this composition. There are contractions $B_{p_j} : \mathcal{H}_{\ell, p_j} \rightarrow \mathcal{H}_{\ell', p_j}$ such that

$$U_{\ell'} F = \left(\bigotimes_{j=1}^s B_{p_j} \right) U_\ell. \quad (1.6)$$

The tensor product is itself a contraction. The list of parties may be chosen to consist exactly of the parties of F . Each map is obtained by composing the local operations at its party. This is the collection of local maps in [1, proof of Theorem 5.2], `04-compression.tex`, lines 233–251.

Proof. ✓ Proof Apply (1.5) successively with $A = \{p_j\}$. The remaining composition has no parties once all p_j have been removed and is the identity on $\mathbb{C}$. The restriction to each party is allowed, so its operator has norm at most one. The tensor product of the resulting contractions gives (1.6). $\square$

Theorem 1.3.6 (Prepared pair sources and local contractions). Lean ✓ Stmt Let P be the prescribed finite set of parties of a gate, and use a fixed ordering of P . Let $F : \mathcal{H}_{\text{in}} \rightarrow \mathcal{H}_{\text{out}}$ be an allowed composition on registers owned by these parties. There is a normalized source list S , containing exactly one source for each unordered source pair occurring in F , and a contraction B_p at each $p \in P$ such that

$$U_{\text{out}} F = \left(\bigotimes_{p \in P} B_p \right) U_{S, \text{in}} P_{S, \mathcal{H}_{\text{in}}}. \quad (1.7)$$

Here $U_{S, \text{in}}$ groups the prepared source registers together with the original input registers by their owners; U_{out} groups the output registers in the same party order. The input and output of B_p are precisely the corresponding memories of party p .

Proof. ✓ Proof Theorem 1.2.5 gives $F = V \circ P_{S, \mathcal{H}_{\text{in}}}$ with V allowed and containing no sources. Every party named by V belongs to the prescribed set P . Apply (1.6) to V with all parties of P , and compose on the right with the preparation map. $\square$

1.4 One-dimensional sources on unused pairs

Definition 1.4.1 (A trivial pair source). Lean Lean Lean Lean ✓ Stmt For distinct parties p, q , the trivial pair source has the complex line at each endpoint and vector $1 \otimes 1 \in \mathbb{C} \otimes \mathbb{C}$. Both endpoint spaces have dimension one, and the source vector is normalized. This is the one-dimensional source assigned to an unused pair in [1, proof of Theorem 5.2], `04-compression.tex`, lines 242–246.

Theorem 1.4.2 (Removing a trivial pair source). Lean Lean Lean ✓ Stmt A trivial pair source can be removed by an allowed composition of local maps containing no source preparations. For any remaining source list S and any original memory $\mathcal{H}$, this composition recovers $P_{S, \mathcal{H}}$ exactly from the preparation with the trivial source placed before S .

Proof. ✓ Proof At each endpoint, use the canonical identification of its one-dimensional register with the empty memory. These maps are isometries and therefore contractions. Applied to the prepared vector $1 \otimes 1$, they leave the remaining memory unchanged. $\square$

Theorem 1.4.3 (A source on every pair). Lean Lean ✓ Stmt Let P be a finite set of parties. Suppose that S is a normalized source list with no repeated unordered pair. There is a list U of sources with one-dimensional endpoint spaces such that the concatenation $T = U ++ S$ is normalized and contains exactly one source for every unordered pair of distinct parties in P .

Every pair of a source in U is absent from S . Thus the original sources are retained with their vectors, spaces, endpoint order, and list order unchanged. For every original memory, an allowed composition containing no sources recovers its preparation by S from its preparation by T .

Proof. ✓ Proof List the unordered pairs of distinct parties that are absent from S . For each such pair, choose either endpoint order and adjoin the corresponding trivial source. The pairs are distinct, and every pair occurs in exactly one of the original and added lists. Repeatedly apply Theorem 1.4.2 to remove the added sources. This also covers empty and singleton party sets, for which no distinct pair exists. $\square$

Theorem 1.4.4 (Local factorization with every pair present). Lean Lean ✓ Stmt Let P be the prescribed finite set of gate parties, with a fixed ordering. Let $F : \mathcal{H}_{\text{in}} \rightarrow \mathcal{H}_{\text{out}}$ be an allowed composition whose register owners and assigned local parties all belong to P . There is a normalized source list T containing exactly one source on every unordered pair of distinct parties in P , and a contraction B_p at each $p \in P$, such that

$$U_{\text{out}} F = \left(\bigotimes_{p \in P} B_p \right) U_{T, \text{in}} P_{T, \mathcal{H}_{\text{in}}}. \quad (1.8)$$

The identifications group registers by party in the prescribed order. The input of B_p consists of that party's original input registers and its halves of the sources in T ; its output consists of that party's original output registers. A source in T whose pair does not occur among the original sources of F has one-dimensional endpoint spaces.

Proof. ✓ Proof First prepare and combine the actual sources of F to obtain $F = V \circ P_{S, \mathcal{H}_{\text{in}}}$, with V allowed and containing no sources. Complete S to T by Theorem 1.4.3. The local maps removing the added sources, followed by V , form another allowed composition containing no sources. Collect its local maps by party using Theorem 1.3.5. $\square$

1.5 Common private source spaces

Definition 1.5.1 (Sources in prescribed pair slots). Lean Lean Lean Lean Lean Lean Lean

Lean ✓ Stmt Fix an ordered list of pairs (p_e, q_e) with $p_e \neq q_e$. Assign complex inner product spaces U_e, V_e to the respective parties and a vector $\eta_e \in U_e \otimes V_e$ to each pair. These data define an ordered source list. Its register layout depends only on the parties and halfspaces, and its unordered pairs are the prescribed pairs. The list is normalized when each η_e is normalized. The empty list prepares no registers; separating the first pair separates its two registers from those of the remaining sources.

For any original memory $\mathcal{H}$, write $Q_{\eta, \mathcal{H}}$ for preparation of these sources, using the fixed register layout and the canonical associativity identifications. Thus $Q_{\eta, \mathcal{H}} x = (\bigotimes_e \eta_e) \otimes x$. These are the pair slots of [1, proof of Theorem 5.2], 04-compression.tex, lines 253–267.

Theorem 1.5.2 (A common order and endpoint orientation). Lean Lean ✓ Stmt Let R be a reference source list with distinct unordered pairs, and let S be a normalized source list with precisely the same distinct unordered pairs. There are normalized sources in the pair order and endpoint orientations of R whose preparation can be transformed into the preparation of S by allowed operations containing no sources, for every spectator memory. Permuting whole source blocks also gives such a transformation.

Proof. ✓ Proof Induct on the reference list. Select the actual source on its first unordered pair and move this source block to the front. If its endpoint order is reversed, apply the tensor-factor exchange $U \otimes V \rightarrow V \otimes U$, which preserves the vector norm. Remove the matched pair and continue. Equal pair sets and distinctness ensure that every source is used exactly once. $\square$

Definition 1.5.3 (Orthogonal sums of branch spaces). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let Ξ be a finite set and let U_ξ be complex inner product spaces. Their common space is the orthogonal sum $\widehat{U} = \bigoplus_{\xi \in \Xi} U_\xi$. Denote its coordinate inclusion by $j_\xi : U_\xi \rightarrow \widehat{U}$ and its coordinate projection by $\pi_\xi : \widehat{U} \rightarrow U_\xi$. The norm is given by $\|(u_\xi)_{\xi \in \Xi}\|^2 = \sum_{\xi \in \Xi} \|u_\xi\|^2$. This is the common-space construction in [1, proof of Theorem 5.2], `04-compression.tex`, lines 253–267.

Theorem 1.5.4 (Embedding and recovering a pair source). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Each j_ξ is isometric, $\|\pi_\xi\| \leq 1$, and $\pi_\xi j_\xi = \mathbb{1}$. For a second family V_ξ , use inclusions k_ξ and projections ρ_ξ on its orthogonal sum. If $\eta_\xi \in U_\xi \otimes V_\xi$, set $\hat{\eta}_\xi = (j_\xi \otimes k_\xi)\eta_\xi$. Then

$$\|\hat{\eta}_\xi\| = \|\eta_\xi\|, \quad (1.9)$$

$$(\pi_\xi \otimes \rho_\xi)\hat{\eta}_\xi = \eta_\xi. \quad (1.10)$$

In particular, normalized sources remain normalized in the common spaces.

Proof. [✓ Proof](#) A coordinate inclusion has exactly one nonzero coordinate, so it preserves norms. Every coordinate norm is bounded by the norm of the orthogonal sum, proving the projection bound. Tensor products of linear isometries preserve norms. The recovery identity follows first on elementary tensors from $\pi_\xi j_\xi = \mathbb{1}$ and $\rho_\xi k_\xi = \mathbb{1}$, then by linearity. $\square$

Theorem 1.5.5 (Local maps on a prepared pair). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $A : U \rightarrow U'$ and $B : V \rightarrow V'$ be continuous linear maps at the respective parties of a pair source. Applying these maps locally, with identities on a spectator memory, sends $u \otimes (v \otimes x)$ to $Au \otimes (Bv \otimes x)$ and satisfies

$$(A \otimes B \otimes \mathbb{1})P_{\eta, \mathcal{H}} = P_{(A \otimes B)\eta, \mathcal{H}}. \quad (1.11)$$

When A and B are contractions, these two local operations form an allowed composition containing no sources.

Proof. [✓ Proof](#) Act first at the first party and then at the second party, keeping the remaining registers untouched. The formula follows on elementary tensors, hence on every source vector by linearity. Tensoring each local contraction with an identity preserves its contraction bound. $\square$

Theorem 1.5.6 (Common spaces for normalized source lists). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $(S_\xi)_{\xi \in \Xi}$ be a finite family of normalized source lists, each with the same distinct unordered pair set as a fixed reference list. There are common halfspaces in the reference pair order and normalized vectors $\hat{\eta}_{\xi, e}$ such that, for every spectator memory $\mathcal{H}$, an allowed composition D_ξ containing no sources satisfies

$$D_\xi Q_{\hat{\eta}_\xi, \mathcal{H}} = P_{S_\xi, \mathcal{H}}. \quad (1.12)$$

The domain of D_ξ is the fixed source register layout together with $\mathcal{H}$; it does not depend on the vectors.

Proof. [✓ Proof](#) Put every list in the reference order using Theorem 1.5.2. At each half of each pair, form the orthogonal sum of its branch spaces and embed the branch vector. Apply the coordinate projections to each pair in turn. Equations (1.10) and (1.11) recover its original source vector. Compose these operations with the exchanges recovering the original source order. All local maps remain contractions. $\square$

Theorem 1.5.7 (Local factorization in common private spaces). Lean Lean ✓ Stmt Let P be the prescribed finite set of gate parties, with a fixed ordering, and let $(F_\xi)_{\xi \in \Xi}$ be a finite family of allowed compositions with the same input and output register layouts. There is one common source slot for every unordered pair of distinct parties, with fixed halfspaces, normalized vectors $\hat{\eta}_{\xi,e}$, and local contractions $B_{\xi,p}$ such that

$$U_{\text{out}} F_\xi = \left(\bigotimes_{p \in P} B_{\xi,p} \right) U_{\text{in}} Q_{\hat{\eta}_\xi, \mathcal{H}_{\text{in}}}. \quad (1.13)$$

Here U_{in} groups the common source registers and the original input registers by party, and U_{out} groups the original output registers in the same party order. Both identifications and the domain and codomain of each party's local maps are independent of ξ . Equivalently, preparation in the common source layout can be followed by an allowed composition containing no sources to recover F_ξ exactly.

These are the common private spaces in [1, proof of Theorem 5.2], `O4-compression.tex`, lines 253–267.

Proof. ✓ Proof Choose a reference ordering of all unordered pairs by completing the empty source list. Complete the actual sources of each F_ξ by Theorem 1.4.4, and then apply Theorem 1.5.6. Compose the resulting recovery maps with the remaining operations of F_ξ and collect the local maps by party. This gives (1.13). The reference list is chosen independently of ξ , so the construction also applies when Ξ is empty. $\square$

Theorem 1.5.8 (The weighted gate in common private spaces). Lean ✓ Stmt Let $F = \sum_{\xi \in \Xi} c_\xi F_\xi$, where Ξ is finite, each $c_\xi \in \mathbb{C}$, and the F_ξ are allowed compositions with the same input and output register layouts on a prescribed finite set of parties P . There are fixed halfspaces for all unordered pairs of distinct parties, normalized pair sources $\hat{\eta}_{\xi,e}$, and local contractions $B_{\xi,p}$ such that

$$U_{\text{out}} F = \sum_{\xi \in \Xi} c_\xi \left(\bigotimes_{p \in P} B_{\xi,p} \right) U_{\text{in}} Q_{\hat{\eta}_\xi, \mathcal{H}_{\text{in}}}. \quad (1.14)$$

The coefficients are those of the original sum. The input and output identifications and all private spaces are common to every summand. This is the common-space form of [1, proof of Theorem 5.2], `O4-compression.tex`, lines 233–267.

Proof. ✓ Proof Apply Theorem 1.5.7 to the family $(F_\xi)_{\xi \in \Xi}$. Multiply each identity (1.13) by c_ξ and sum over Ξ . $\square$

Theorem 1.5.9 (Finite coordinates for a finite family of source vectors). Lean ✓ Stmt Let U, V be complex inner product spaces, let Ξ be finite, and let $\eta_\xi \in U \otimes V$ for each $\xi \in \Xi$, where the tensor product is algebraic. There exist $a, b \in \mathbb{N}$, linear isometries $f : \mathbb{C}^a \rightarrow U$ and $g : \mathbb{C}^b \rightarrow V$, and vectors $\eta_\xi^0 \in \mathbb{C}^a \otimes \mathbb{C}^b$ such that $(f \otimes g) \eta_\xi^0 = \eta_\xi$ and $\|\eta_\xi^0\| = \|\eta_\xi\|$ for every ξ . The integers a, b may be zero. This gives finite coordinates for the source vectors used in the Schmidt decompositions of [1, proof of Theorem 5.2], `O4-compression.tex`, lines 279–299.

Proof. ✓ Proof A finite set of algebraic tensors is contained in $U_0 \otimes V_0$ for some finite-dimensional subspaces $U_0 \subseteq U$ and $V_0 \subseteq V$: express each tensor as a finite sum of elementary tensors and span the finitely many factors on each side. Choose orthonormal coordinates on U_0, V_0 and compose their coordinate isometries with the subspace inclusions. The induced tensor map is isometric, so the coordinate vectors have exactly the original norms. $\square$

Theorem 1.5.10 (Source preparations in finite coordinates). [Lean](#) [Lean](#) [✓ Stmt](#) Fix a finite source list, common halfspaces U_e, V_e , and a finite family of vectors $\eta_{\xi,e} \in U_e \otimes V_e$. There are dimensions a_e, b_e , isometries $f_e : \mathbb{C}^{a_e} \rightarrow U_e$ and $g_e : \mathbb{C}^{b_e} \rightarrow V_e$, and coordinate vectors $\eta_{\xi,e}^0$ such that $(f_e \otimes g_e)\eta_{\xi,e}^0 = \eta_{\xi,e}$ and $\|\eta_{\xi,e}^0\| = \|\eta_{\xi,e}\|$. For every ξ and every spectator memory $\mathcal{H}$, an allowed composition D_ξ containing no sources satisfies $D_\xi Q_{\eta_{\xi,e}^0, \mathcal{H}} = Q_{\eta_{\xi,e}, \mathcal{H}}$. More generally, any prescribed isometric inclusions of the two halves of each source induce such a recovery of the included sources.

Proof. [✓ Proof](#) Apply Theorem 1.5.9 at each pair slot. Apply f_e and g_e locally to the two halves of that slot, then proceed through the source list. Equation (1.11) recovers the corresponding source vector at each step. The inclusions are isometries, hence contractions, and create no further sources. $\square$

Theorem 1.5.11 (The source-gate expansion in finite coordinates). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $F = \sum_{\xi \in \Xi} c_\xi F_\xi$, where Ξ is finite, $c_\xi \in \mathbb{C}$, and each F_ξ is an allowed composition between fixed register layouts on a prescribed finite set of parties P . There are common dimensions a_e, b_e for every unordered pair of distinct parties, normalized sources $\eta_{\xi,e}^0 \in \mathbb{C}^{a_e} \otimes \mathbb{C}^{b_e}$, and local contractions $B_{\xi,p}$ such that

$$U_{\text{out}} F_\xi = \left(\bigotimes_{p \in P} B_{\xi,p} \right) U_{\text{in}} Q_{\eta_{\xi,e}^0, \mathcal{H}_{\text{in}}}, \quad (1.15)$$

$$U_{\text{out}} F = \sum_{\xi \in \Xi} c_\xi \left(\bigotimes_{p \in P} B_{\xi,p} \right) U_{\text{in}} Q_{\eta_{\xi,e}^0, \mathcal{H}_{\text{in}}}. \quad (1.16)$$

Here U_{in} groups the common finite source registers and original input registers by party, while U_{out} groups the unchanged output registers in the same order. Each branch is also the exact composition of its coordinate source preparation with allowed operations containing no sources.

This is an exact finite-coordinate form of [1, proof of Theorem 5.2], `04-compression.tex`, lines 233–299. No bound on the private dimensions a_e, b_e is asserted.

Proof. [✓ Proof](#) Complete each branch's source list and put all lists in common spaces. Apply Theorem 1.5.10 to their finitely many vectors at each slot. Compose the coordinate inclusions with the recovery of each branch's actual sources and its remaining operations. All these operations are local contractions. Collect them by party and then sum the resulting branch identities with the original coefficients c_ξ . $\square$

1.6 Density expansions of source gates

Fix the ordered pair slots of Definition 1.5.1, with halfspaces U_e, V_e , and write $\mathcal{S}$ for their joint register space. All tensor products below use this order and its canonical associativity identifications. The slot set and all finite index sets may be empty. These are the source-coordinate identities used in [1, proof of Theorem 5.2], `04-compression.tex`, lines 279–355.

Definition 1.6.1 (The joint source vector). [Lean](#) [Lean](#) [✓ Stmt](#) The joint source vector is the multilinear map $\Phi : \prod_e (U_e \otimes V_e) \rightarrow \mathcal{S}$ defined recursively by $\Phi_\emptyset = 1 \in \mathbb{C}$ and $\Phi_{e::E}(\eta) = \eta_e \otimes \Phi_E(\eta|_E)$, with the first pair separated into its two registers. Thus $\Phi(\eta)$ is the ordered tensor product of the pair vectors, including the terminal scalar factor.

Theorem 1.6.2 (Preparation and finite sums of sources). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For every original memory $\mathcal{H}$, the actual source preparation satisfies $Q_{\eta, \mathcal{H}} x = \Phi(\eta) \otimes x$. In particular, let J_e be finite, let $d_e : J_e \rightarrow \mathbb{C}$, and let $\theta_{e,j} \in U_e \otimes V_e$. Then

$$Q_{(\sum_{j \in J_e} d_e(j) \theta_{e,j})_e, \mathcal{H}} = \sum_{a \in \prod_e J_e} \left(\prod_e d_e(a_e) \right) Q_{(\theta_{e,a_e})_e, \mathcal{H}}. \quad (1.17)$$

Proof. [✓ Proof](#) Induction on the source list identifies $\Phi(\eta)$ with its actual joint vector: the empty list contributes 1, and the induction step prepares η_e in front of the remaining sources. Changing to the fixed register layout preserves this identity and its tensor product with x . Expand each argument of the multilinear map Φ as a finite sum. Its value on a scalar multiple in every argument is $\Phi((d_e \theta_e)_e) = (\prod_e d_e) \Phi((\theta_e)_e)$. Tensoring with x gives (1.17). $\square$

Definition 1.6.3 (Matrices of prepared branches). [Lean](#) [✓ Stmt](#) Let $T : \mathcal{S} \otimes \mathcal{H}_{\text{in}} \rightarrow \mathcal{H}_{\text{out}}$ be the operator of a source-only composition as in Definition 1.1.13, without imposing normalization or contraction bounds. Choose finite orthonormal bases f_j and g_i of the input and output memories. Denote the matrix of the prepared branch by

$$K_T(\eta)_{ij} = \langle g_i, T Q_{\eta, \mathcal{H}_{\text{in}}} f_j \rangle. \quad (1.18)$$

The input and output bases are fixed throughout the density expansions below. No finite-dimensionality assumption on the private spaces is required for this definition.

Theorem 1.6.4 (Coordinates of source preparation). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For finite families as in Theorem 1.6.2,

$$K_T((\sum_j d_e(j) \theta_{e,j})_e) = \sum_{a \in \prod_e J_e} \left(\prod_e d_e(a_e) \right) K_T((\theta_{e,a_e})_e). \quad (1.19)$$

In particular, choose finite orthonormal bases $(u_{e,r})_r$ of U_e and $(v_{e,s})_s$ of V_e , and put $\eta_e(r, s) = \langle u_{e,r} \otimes v_{e,s}, \eta_e \rangle$. For $a_e = (r_e, s_e)$, set $\theta_{e,a_e} = u_{e,r_e} \otimes v_{e,s_e}$. Formula (1.19) then holds with $d_e(a_e) = \eta_e(a_e)$ and left-hand side $K_T(\eta)$. The identical expansion holds for $Q_{\eta, \mathcal{H}}$.

Proof. [✓ Proof](#) Compose (1.17) with T and take matrices in the prescribed bases. Both operations are linear. For the basis specialization, use $\eta_e = \sum_{r,s} \eta_e(r, s) u_{e,r} \otimes v_{e,s}$ in each slot. $\square$

Definition 1.6.5 (Density coefficients of actual branches). [Lean](#) [✓ Stmt](#) Choose vector families $u_{e,r} \in U_e$, $v_{e,s} \in V_e$ for a ket branch T , and independent families $\tilde{u}_{e,r} \in U_e$, $\tilde{v}_{e,s} \in V_e$ for a bra branch T' . Their index sets may differ. Write $K_T(a) = K_T((u_{e,a_e^{(1)}} \otimes v_{e,a_e^{(2)}})_e)$ and define $K_{T'}(b)$ using the second families. For any complex input matrix ρ , define

$$C_{T,T'}(\rho; a, b) = K_T(a) \rho K_{T'}(b)^\dagger. \quad (1.20)$$

The families need not span the private spaces or be orthogonal.

Theorem 1.6.6 (Density expansion for finite component families). [Lean](#) [✓ Stmt](#) Suppose the families in Definition 1.6.5 have finite index sets. Choose complex coefficients $d_e(r, s)$ and $\tilde{d}_e(r, s)$, and put

$$\eta_e = \sum_{r,s} d_e(r, s) u_{e,r} \otimes v_{e,s}, \quad (1.21)$$

$$\zeta_e = \sum_{r,s} \tilde{d}_e(r, s) \tilde{u}_{e,r} \otimes \tilde{v}_{e,s}. \quad (1.22)$$

For every input matrix ρ ,

$$K_T(\eta)\rho K_{T'}(\zeta)^\dagger = \sum_{a,b} \left(\prod_e d_e(a_e) \overline{\tilde{d}_e(b_e)} \right) C_{T,T'}(\rho; a, b). \quad (1.23)$$

This identity also applies to vector families contained in proper subspaces of the private halfspaces.

Proof. ✓ Proof Expand the two prepared matrices using (1.19). The adjoint conjugates each scalar coefficient on the bra side. Distributing both finite sums yields (1.23), since

$$\left(\prod_e d_e(a_e) \right) \overline{\prod_e \tilde{d}_e(b_e)} = \prod_e d_e(a_e) \overline{\tilde{d}_e(b_e)}.$$

□

Theorem 1.6.7 (Coordinate density expansion of a weighted gate). Lean Lean ✓ Stmt Let Ξ be finite. For each $\xi \in \Xi$, choose a branch T_ξ , source vectors $\eta_{\xi,e}$, and finite orthonormal bases $(u_{\xi,e,r})_r, (v_{\xi,e,s})_s$ of the two halfspaces. The bases and their index sets may depend on ξ . Set $K_\xi = K_{T_\xi}(\eta_\xi)$ and $\kappa_\xi(a) = \prod_e \langle u_{\xi,e,a_e^{(1)}} \otimes v_{\xi,e,a_e^{(2)}}, \eta_{\xi,e} \rangle$. For any input matrix ρ and any two branches,

$$K_\xi \rho K_\zeta^\dagger = \sum_{a,b} \kappa_\xi(a) \overline{\kappa_\zeta(b)} C_{T_\xi, T_\zeta}(\rho; a, b). \quad (1.24)$$

If $G = \sum_\xi c_\xi K_\xi$, then

$$G \rho G^\dagger = \sum_{\xi, \zeta} c_\xi \overline{c_\zeta} \sum_{a,b} \kappa_\xi(a) \overline{\kappa_\zeta(b)} C_{T_\xi, T_\zeta}(\rho; a, b). \quad (1.25)$$

Neither identity requires positivity or normalization of ρ .

Proof. ✓ Proof Expand each source in its branch's endpoint bases and distribute the ket and bra sums, conjugating only the bra coefficients. This gives (1.24). Expand $G \rho G^\dagger = \sum_{\xi, \zeta} c_\xi \overline{c_\zeta} K_\xi \rho K_\zeta^\dagger$ and substitute (1.24) to obtain (1.25). □

Theorem 1.6.8 (Operator norm of a prepared branch). Lean Lean ✓ Stmt If T is an allowed composition and $\|\eta_e\| = 1$ in every slot, then $\|K_T(\eta)\| \leq 1$ in the Euclidean operator norm. In particular, if every $u_{e,r}$ and $v_{e,s}$ has norm one, then $\|K_T(a)\| \leq 1$ for every coordinate assignment a . The unit-vector families need not span their halfspaces. These are the coefficient contractions in [1, proof of Theorem 5.2], `O4-compression.tex`, lines 409–427.

Proof. ✓ Proof The normalization of each source makes its preparation allowed, hence contractive. Composing with T gives $\|TQ_{\eta, \mathcal{H}_{\text{in}}}\| \leq 1$. Orthonormal coordinate maps are isometries, so the matrix in (1.18) has the same operator norm. For the last assertion, use $\|u_{e,r} \otimes v_{e,s}\| = \|u_{e,r}\| \|v_{e,s}\| = 1$. □

Theorem 1.6.9 (Density coefficients constructed from a gate). Lean ✓ Stmt Let P and Ξ be finite. For every $\xi \in \Xi$, let F_ξ be an allowed source-only composition on the parties P , with fixed input and output memories. Fix finite orthonormal bases of these memories and complex coefficients c_ξ , and let M_ξ be the matrix of F_ξ in these bases.

There is one ordered slot for each unordered pair of distinct parties, with common finite halfspaces $U_e = \mathbb{C}^{d_e}$ and $V_e = \mathbb{C}^{d'_e}$, normalized vectors $\eta_{\xi,e} \in U_e \otimes V_e$, and allowed compositions T_ξ containing no sources such that $F_\xi = T_\xi Q_{\eta_\xi, \mathcal{H}_{\text{in}}}$. Use the standard orthonormal bases of these

halfspaces to define $K_\xi(a)$ as in Definition 1.6.5, and put $\kappa_\xi(a) = \prod_e \eta_{\xi,e}(a_e)$. Each $K_\xi(a)$ has Euclidean operator norm at most one.

These choices are independent of the input matrix ρ . For every complex input matrix ρ , writing $G = \sum_\xi c_\xi M_\xi$, one has

$$G\rho G^\dagger = \sum_{\xi,\zeta} c_\xi \overline{c_\zeta} \sum_{a,b} \kappa_\xi(a) \overline{\kappa_\zeta(b)} K_\xi(a) \rho K_\zeta(b)^\dagger. \quad (1.26)$$

The halfspaces, source vectors, and coefficient matrices are obtained from the original compositions. This is the exact source-coordinate density identity preceding the random replacement in [1, proof of Theorem 5.2], `04-compression.tex`, lines 233–355.

Proof. ✓ Proof Theorem 1.5.11 constructs the common finite source spaces, their normalized vectors, and the exact source-free compositions T_ξ . Replacing every source half by a standard basis vector gives normalized elementary preparations; Theorem 1.6.8 therefore proves $\|K_\xi(a)\| \leq 1$. Taking matrices of $F_\xi = T_\xi Q_{\eta_\xi, \mathcal{H}_{\text{in}}}$ identifies M_ξ with the prepared branch matrix. Apply Theorem 1.6.7 to obtain (1.26) for every ρ . □

1.7 Joint contractions of the free source inputs

Fix ordered source slots with halfspaces U_e, V_e and joint register space $\mathcal{S}$, as in Definition 1.5.1. All source and physical input coordinates in this section remain free. This gives the contractions used in [1, proof of Theorem 5.2], `04-compression.tex`, lines 383–468.

Definition 1.7.1 (Bases of source and input registers). Lean Lean Lean Lean ✓ Stmt Choose finite orthonormal bases $(u_{e,r})_{r \in I_e}$ of U_e and $(v_{e,s})_{s \in J_e}$ of V_e . Let $\mathcal{A} = \prod_e (I_e \times J_e)$. The orthonormal basis of $\mathcal{S}$ indexed by $a \in \mathcal{A}$ consists of

$$s_a = \Phi((u_{e,a_e(1)} \otimes v_{e,a_e(2)})_e). \quad (1.27)$$

The empty source list has its single basis vector $1 \in \mathbb{C}$. If $(f_j)_{j \in N}$ is an orthonormal basis of the original input memory $\mathcal{H}$, the resulting orthonormal basis of $\mathcal{S} \otimes \mathcal{H}$ is indexed by (a, j) and satisfies

$$s_a \otimes f_j = Q_{(u_{e,a_e(1)} \otimes v_{e,a_e(2)})_e, \mathcal{H}} f_j. \quad (1.28)$$

Both formulas use the canonical associativity identifications.

Proof. ✓ Proof Induct on the ordered slots. Tensor the bases of the first two halfspaces with the basis already constructed for the remaining slots, and identify a full assignment with its first component and its restriction to the remaining slots. Tensor products and isometric identifications preserve orthonormal bases. This gives (1.27); tensoring with the input basis and applying the preparation identity gives (1.28). □

Definition 1.7.2 (The complete free-input matrix). Lean Lean ✓ Stmt Let $T : \mathcal{S} \otimes \mathcal{H} \rightarrow \mathcal{K}$ be the operator of a composition, and choose an orthonormal output basis $(g_k)_{k \in M}$. Its complete matrix A_T has row index $k \in M$ and column index $(a, j) \in \mathcal{A} \times N$:

$$A_T(k, (a, j)) = \langle g_k, T(s_a \otimes f_j) \rangle = K_T(a)_{kj}. \quad (1.29)$$

Here $K_T(a)$ is the actual prepared matrix of Definition 1.6.5.

Theorem 1.7.3 (Joint contraction of all source inputs). [Lean](#) [✓ Stmt](#) If T is an allowed composition, then $\|A_T\| \leq 1$ in the Euclidean operator norm. This bounds its action on arbitrary superpositions of the source and physical input coordinates, independently of the number of slots and their dimensions.

Proof. [✓ Proof](#) The complete matrix is the matrix of the same operator T in orthonormal input and output bases. Coordinate maps preserve norms, so $\|A_T\| = \|T\| \leq 1$. $\square$

Theorem 1.7.4 (The bra-adjoint-ket contraction). [Lean](#) [Lean](#) [✓ Stmt](#) Let T and T' be compositions with a common output memory and output basis. Their source lists, source spaces, source bases, and physical input memories and bases may differ. Write $A = A_T$ and $B = A_{T'}$, and let a, b denote the respective source assignments. Then the rectangular matrix $O = B^\dagger A$ satisfies

$$O_{(b,j),(a,i)} = (K_{T'}(b)^\dagger K_T(a))_{ji}. \quad (1.30)$$

Thus the bra source assignment and its physical input index form the row index of O . If both compositions are allowed, then $\|O\| \leq 1$.

Proof. [✓ Proof](#) Multiplying the matrices in (1.29) gives (1.30). The norm bound follows from $\|B^\dagger A\| \leq \|B^\dagger\| \|A\| = \|B\| \|A\| \leq 1$. $\square$

Theorem 1.7.5 (Tracing a prepared density coefficient). [Lean](#) [Lean](#) [✓ Stmt](#) With the ket and bra input coordinates as above, let $\rho \in \mathbb{C}^{N \times N'}$ be any rectangular matrix. No contraction assumption on T or T' is needed for the identity

$$\text{tr}(K_T(a) \rho K_{T'}(b)^\dagger) = \sum_{i \in N} \sum_{j \in N'} \rho_{ij} O_{(b,j),(a,i)}. \quad (1.31)$$

When the two branches have common source spaces and physical input coordinates, this is the trace of $C_{T,T'}(\rho; a, b)$ from Definition 1.6.5, with independent endpoint bases on the two sides.

Proof. [✓ Proof](#) Cyclicity of the rectangular trace gives $\text{tr}(K_T(a) \rho K_{T'}(b)^\dagger) = \text{tr}(\rho K_{T'}(b)^\dagger K_T(a))$. Expand the trace and substitute the block identity (1.30). $\square$

Definition 1.7.6 (Pair labels of the actual slots). [Lean](#) [Lean](#) [✓ Stmt](#) Let R be a source list whose unordered pair labels are distinct and contain every unordered pair of distinct parties exactly once. Sending a slot to the unordered pair of its actual source is a bijection from its slot indices to $\{\{p, q\} : p, q \in P, p \neq q\}$. This identification depends only on the parties of each source.

Theorem 1.7.7 (Number of pair-source slots). [Lean](#) [Lean](#) [✓ Stmt](#) If P is finite and R has the pair coverage and distinctness in Definition 1.7.6, then $|R| = \binom{|P|}{2} \leq |P|^2$. This includes empty and singleton party sets. These are the pair slots used in [1, proof of Theorem 5.2], *04-compression.tex*, lines 565–585.

Proof. [✓ Proof](#) Count the unordered pairs of distinct elements under the stated bijection. Their number is $\binom{|P|}{2}$, which is at most $|P|^2$. $\square$

Definition 1.7.8 (Local maps on source slots). [Lean](#) [✓ Stmt](#) For each slot e , let $L_e : U_e \rightarrow U'_e$ and $M_e : V_e \rightarrow V'_e$ be continuous linear maps. Apply these maps to the corresponding source registers and leave the original input memory $\mathcal{H}$ unchanged. Denote the resulting composition by $F_{L,M}$. This composition depends on the maps and the ordered slots, but not on the prepared source vectors.

Theorem 1.7.9 (Action of the local source maps). [Lean](#) [Lean](#) [✓ Stmt](#) For every family $\eta_e \in U_e \otimes V_e$,

$$F_{L,M} Q_{\eta, \mathcal{H}} = Q_{((L_e \otimes M_e) \eta_e)_e, \mathcal{H}}. \quad (1.32)$$

If $\|L_e\|, \|M_e\| \leq 1$ for every slot, the composition $F_{L,M}$ is allowed and contains no source preparations. The identity requires neither these norm bounds nor normalization of the vectors η_e .

Proof. [✓ Proof](#) Induct on the ordered source slots. Apply the two endpoint maps of the first source, and then apply the construction to the remaining sources while leaving those first two registers fixed. Moving the remaining preparation past these fixed registers gives (1.32). Each constituent is a local map, and its allowedness follows from the corresponding endpoint norm bound. $\square$

Definition 1.7.10 (The complete matrix in source frames). [Lean](#) [Lean](#) [✓ Stmt](#) Let $T : \mathcal{S}' \otimes \mathcal{H} \rightarrow \mathcal{K}$ be a composition on the target halfspaces U'_e, V'_e . Use the finite orthonormal bases of U_e, V_e and the physical input and output bases already specified. Define

$$A_T^{L,M}(k, (a, j)) = \left\langle g_k, T Q_{(L_e u_{e, a_e(1)} \otimes M_e v_{e, a_e(2)})_e, \mathcal{H}} f_j \right\rangle. \quad (1.33)$$

This is the complete free-input matrix of $TF_{L,M}$. The images of the endpoint maps need not span the target halfspaces.

Proof. [✓ Proof](#) Evaluate (1.32) on each elementary endpoint-basis preparation and each physical input basis vector. The resulting columns agree with (1.33). $\square$

Theorem 1.7.11 (Joint contraction in source frames). [Lean](#) [✓ Stmt](#) Assume that T is allowed and that each L_e and M_e is a contraction. Then $\|A_T^{L,M}\| \leq 1$ in the Euclidean operator norm. In particular, this applies to isometric inclusions of proper Schmidt subspaces. No orthonormal basis of the target halfspaces is required.

Proof. [✓ Proof](#) The composition $TF_{L,M}$ is allowed. Apply the joint contraction bound to its complete free-input matrix. $\square$

1.8 Grouping parties into two sides

Let $f : P \rightarrow Q$ be an arbitrary map of party labels. We group all parties with the same image while retaining every register and every operation. This is the owner identification used for the affected and exterior systems in [1, proof of Theorem 5.2], `04-compression.tex`, lines 383–427.

Definition 1.8.1 (Changing register owners). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) If $\ell = ((p_1, H_1), \dots, (p_n, H_n))$ is a register list, let $\ell^f = ((f(p_1), H_1), \dots, (f(p_n), H_n))$. Empty lists and concatenation are preserved. The canonical isometry $R_{f, \ell} : \mathcal{H}_\ell \rightarrow \mathcal{H}_{\ell^f}$ is the tensor product of the identities on the original register spaces. On the empty list it is the identity of $\mathbb{C}$, and on an elementary tensor it satisfies

$$R_{f, (p, H)\ell}(x \otimes y) = x \otimes R_{f, \ell} y. \quad (1.34)$$

No injectivity or finiteness assumption on f , P , or Q is imposed.

Theorem 1.8.8 (Exact preservation of an allowed composition). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#)
The operators of F and F^f satisfy

$$R_{f,\ell'} F = F^f R_{f,\ell}. \quad (1.36)$$

This identity does not require norm bounds. If F is allowed, then F^f is allowed. Its pair-source list is exactly S_f , where S is the pair-source list of F .

Proof. [Stmnt](#) Induct on the chronological composition. For a local map, (1.35) moves the owner identifications through its local and spectator factors; the conjugating isometries cancel. An internal source has the same preparation operator by Theorem 1.8.6, while a surviving source has the same vector and endpoint spaces. Exchanges and operations beside untouched registers commute with the tensor product of identities. Composition preserves the equality. Isometric conjugation preserves each local contraction bound, and an internal normalized source is an allowed local preparation. The description of the source list follows by the same induction. $\square$

Theorem 1.8.9 (Two contractions when every source is internal). [Lean](#) [Stmnt](#) Let $f : P \rightarrow \{0, 1\}$, and suppose F is allowed. Assume that $f(p) = f(q)$ for the endpoints of every pair source occurring in F . Let $\mathcal{H}_{\ell,+}$ and $\mathcal{H}_{\ell,-}$ be the memories of the registers with owner images 1 and 0, respectively, in their original relative order. Let $J_{f,\ell} : \mathcal{H}_\ell \rightarrow \mathcal{H}_{\ell,+} \otimes \mathcal{H}_{\ell,-}$ first change the owner labels and then partition the two owners. There are two allowed compositions A_+, A_- on the respective memories, each containing no pair sources and each having operator norm at most one, such that

$$J_{f,\ell'} F = (A_+ \otimes A_-) J_{f,\ell}. \quad (1.37)$$

The two compositions are the actual restrictions of F^f to the two owners.

Proof. [Stmnt](#) The internal-source condition makes S_f empty. Hence F^f contains no pair preparations, while remaining an allowed composition. Apply Theorem 1.3.4 to its two owners and compose the resulting identity with (1.36). Its two restricted compositions are allowed and therefore have norm at most one. $\square$

1.9 Free source inputs and untouched exterior gates

The separation argument in [1, proof of Theorem 5.2], 04-compression.tex, lines 342–434, uses corrected source registers as free inputs. Exact sources internal to either side are prepared there, while gates wholly in the exterior retain their complete operators. We give the corresponding exact constructions.

1.9.1 Finite gate expansions

Definition 1.9.1.1 (A gate with common finite source positions). [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Fix a finite branch set Ξ , coefficients $c_\xi \in \mathbb{C}$, and input and output register lists a, b on the gate's participating party set P . Common finite source data consist of an ordered list E containing each unordered pair of distinct parties exactly once, spaces $U_e = \mathbb{C}^{u_e}$ and $V_e = \mathbb{C}^{v_e}$, normalized vectors $\eta_{\xi,e} \in U_e \otimes V_e$, and allowed compositions $T_\xi : \mathcal{S} \otimes \mathcal{H}_a \rightarrow \mathcal{H}_b$ containing no pair preparations. Here $\mathcal{S}$ is the memory of all source registers in their prescribed order. Write $F_\xi = T_\xi Q_{\eta_\xi, \mathcal{H}_a}$ and

$$G = \sum_{\xi \in \Xi} c_\xi F_\xi, \quad (1.38)$$

and require $\|G\| \leq 1$. The coefficients are fixed before these data are chosen. Completion by unused pairs is taken only over the participating parties P of this gate.

Theorem 1.9.1.2 (Prepared branches and the complete gate). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Each F_ξ is an allowed composition. Its ordered source list is precisely the list of prescribed pairs with spaces U_e, V_e and vectors $\eta_{\xi,e}$. The complete sum G , with its original scalar coefficients, is a contraction.*

Proof. [✓ Proof](#) Normalization makes the source preparation allowed. Compose it with T_ξ , whose source list is empty. Thus the only sources are the prescribed preparations. The bound on G is part of the gate data. $\square$

Theorem 1.9.1.3 (Construction from the original branch operators). [Lean](#) [✓ Stmt](#) *Suppose P and Ξ are finite, each $W_\xi : a \rightarrow b$ is an allowed composition, and $\|\sum_{\xi \in \Xi} c_\xi W_\xi\| \leq 1$. There are common finite source data with the same coefficients c_ξ for which $G = \sum_{\xi \in \Xi} c_\xi W_\xi$ exactly. No nonemptiness assumption on Ξ is needed.*

Proof. [✓ Proof](#) Apply Theorem 1.5.11 to the original branch compositions. It gives common finite spaces, normalized vectors, and allowed compositions without sources, with $T_\xi Q_{\eta_\xi, \mathcal{H}_a} = W_\xi$ for each ξ . Substitution in (1.38) preserves the original sum and therefore its contraction bound. $\square$

1.9.2 Affected parties and the exterior

Definition 1.9.2.1 (Keeping affected owners distinct). [Lean](#) [Lean](#) [✓ Stmt](#) Let $\mathcal{A} \subseteq P$ have decidable membership. Set $Q = \mathcal{A} \sqcup \{*\}$, where $*$ denotes the exterior, and define $g : P \rightarrow Q$ by

$$g(p) = \begin{cases} p, & p \in \mathcal{A}, \\ *, & p \notin \mathcal{A}. \end{cases} \quad (1.39)$$

Thus $g(p) = *$ exactly when p is exterior.

Theorem 1.9.2.2 (Equality of grouped owners). [Lean](#) [Lean](#) [✓ Stmt](#) *One has $g(p) = g(q)$ exactly when $p = q$ or both parties are exterior. Consequently, for distinct source endpoints p, q , $g(p) \neq g(q)$ exactly when at least one endpoint belongs to $\mathcal{A}$. Both corrected pairs within $\mathcal{A}$ and pairs crossing to its exterior remain between distinct owners.*

Proof. [✓ Proof](#) Separate the four possibilities for membership of p, q in $\mathcal{A}$. Two affected images agree only when their original parties agree; an affected image cannot equal $*$. $\square$

Theorem 1.9.2.3 (The surviving ordered source occurrences). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *A source occurs in S_g precisely when it is obtained from an occurrence $s = (p, q; U, V; \eta)$ in S with at least one affected endpoint, by replacing its owners with $g(p), g(q)$. Its spaces and vector are unchanged. The ordered pair labels are obtained by filtering S by this same condition and then applying g to both endpoints. Order and multiplicity are retained, even when different original pairs acquire the same image. In particular, S_g is empty exactly when both endpoints of every original source are exterior.*

Proof. [✓ Proof](#) Substitute the distinct-owner criterion of Theorem 1.9.2.2 into the occurrence description of S_g . Induction on the original source list gives the ordered pair-label identity. The empty-list criterion follows from Theorem 1.8.4. $\square$

Definition 1.9.2.4 (Retaining a complete operation at one grouped owner). [Lean](#) [✓ Stmt](#) Let $f : P \rightarrow Q$, and suppose every register in a and b has owner image $q \in Q$. For $D : \mathcal{H}_a \rightarrow \mathcal{H}_b$, regard $R_{f,b} D R_{f,a}^{-1}$ as a local operation at q and leave the spectator list ℓ untouched. This gives an actual composition $D_\ell^f : (a\ell)^f \rightarrow (b\ell)^f$.

Theorem 1.9.2.5 (Exact action of an untouched aggregate). [Lean](#) [Lean](#) [✓ Stmt](#) The complete operation satisfies

$$D_\ell^f R_{f,a\ell} = R_{f,b\ell} C_{b,\ell}^{-1} (D \otimes I_{\mathcal{H}_\ell}) C_{a,\ell}. \quad (1.40)$$

If $\|D\| \leq 1$, this composition is allowed and contains no pair sources. For $f = g$ and $q = *$, this retains an exterior gate as its original aggregate contraction, without expanding its branch sum.

Proof. [✓ Proof](#) Apply the concatenation identity for the canonical owner isometries. The two isometries conjugating D cancel with the corresponding input and output identifications, giving (1.40). Conjugation by isometries preserves the contraction bound. A local operation creates no pair sources. $\square$

Theorem 1.9.2.6 (Grouping preserves the weighted gate sum). [Lean](#) [✓ Stmt](#) For a finite family of compositions $F_\xi : a \rightarrow b$ and arbitrary coefficients $c_\xi \in \mathbb{C}$,

$$\left(\sum_\xi c_\xi F_\xi^f \right) R_{f,a} = R_{f,b} \left(\sum_\xi c_\xi F_\xi \right). \quad (1.41)$$

Proof. [✓ Proof](#) Sum the exact identity $F_\xi^f R_{f,a} = R_{f,b} F_\xi$ with coefficients c_ξ . Composition is linear on both sides. $\square$

Definition 1.9.2.7 (The complete gate after grouping). [Lean](#) [✓ Stmt](#) For common finite source data and any map $f : P \rightarrow Q$, define the grouped aggregate by $G^f = R_{f,b} G R_{f,a}^{-1}$.

Theorem 1.9.2.8 (The original expansion and aggregate norm after grouping). [Lean](#) [Lean](#) [✓ Stmt](#) The exact branch expansion is $G^f = \sum_\xi c_\xi F_\xi^f$, and $\|G^f\| \leq 1$. This bound comes from the original aggregate contraction and does not involve $\sum_\xi |c_\xi|$.

Proof. [✓ Proof](#) Compose (1.41) with $R_{f,a}^{-1}$. Both canonical owner maps are isometries, so conjugating the aggregate preserves its norm bound. $\square$

1.9.3 Preparing only the fixed sources

Definition 1.9.3.1 (The selected original source positions). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let E be an ordered list of pair positions, with halfspaces U_e, V_e , and let $E_{\text{free}} \subseteq E$ be selected. The selected source list retains these occurrences in their original order, with their original endpoint parties and the prescribed spaces and vectors. Its register layout is independent of those vectors. Denote its memory by $\mathcal{S}_{\text{free}}$ and the memory of all positions by $\mathcal{S}$.

Definition 1.9.3.2 (Selective source preparation). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Fix vectors $\eta_e \in U_e \otimes V_e$ only at positions outside E_{free} . Define $\Phi_\eta : \mathcal{S}_{\text{free}} \otimes \mathcal{H}_\ell \rightarrow \mathcal{S} \otimes \mathcal{H}_\ell$ by preparing the fixed sources and retaining all free registers in their original positions. This is one actual composition, independent of the vectors later supplied to the free registers.

For an arbitrary family $\zeta_e \in U_e \otimes V_e$, write $Q_{\zeta, \mathcal{H}_\ell}^{\text{free}}$ for preparation at just the free positions. The completed family θ has $\theta_e = \zeta_e$ at free positions and $\theta_e = \eta_e$ elsewhere.

Theorem 1.9.3.3 (Only the fixed sources are prepared). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The source list of Φ_η is exactly the list of fixed original positions, in their original order, with vectors η_e . In particular, each prepared source has the original endpoint parties and halfspaces of its fixed position. If every fixed vector has norm one, then Φ_η is allowed.*

Proof. [✓ Proof](#) Induct on the source list. At a free position, retain the two registers and apply the construction to the remaining positions. At a fixed position, prepare its source after the remaining positions. The source inventory records precisely this fixed occurrence. Its preparation is allowed by normalization. $\square$

Theorem 1.9.3.4 (Uniform recovery of every completed preparation). [Lean](#) [✓ Stmt](#) *For every family of free vectors, without a normalization assumption,*

$$\Phi_\eta Q_{\zeta, \mathcal{H}_\ell}^{\text{free}} = Q_{\theta, \mathcal{H}_\ell}. \quad (1.42)$$

Proof. [✓ Proof](#) Induct on the ordered positions. At a free head, preparation of its vector commutes with the operations on the remaining registers. At a fixed head, both sides prepare the same fixed vector. In each case the induction hypothesis gives the equality for the remaining positions. The empty source list gives the identity on the original input memory. $\square$

Theorem 1.9.3.5 (Two contractions independent of the free source vectors). [Lean](#) [✓ Stmt](#) *Let $f : P \rightarrow \{0, 1\}$ specify two sides. Suppose every fixed source has both endpoints on the same side and has norm one. Let $T : \mathcal{S} \otimes \mathcal{H}_\ell \rightarrow \mathcal{H}_{\ell'}$ be an allowed composition containing no pair sources. There are two allowed compositions A_+, A_- , containing no pair sources and each having norm at most one, such that*

$$J_{f, \ell'} T \Phi_\eta = (A_+ \otimes A_-) J_{f, L}, \quad (1.43)$$

$$J_{f, \ell'} T Q_{\theta, \mathcal{H}_\ell} = (A_+ \otimes A_-) J_{f, L} Q_{\zeta, \mathcal{H}_\ell}^{\text{free}}. \quad (1.44)$$

Here L is the free-source register list followed by ℓ , and $J_{f, L}$ is its canonical partition isometry. The same two compositions work for every family ζ of free vectors. Thus the contraction bounds apply on all free inputs.

Proof. [✓ Proof](#) The only sources in $T \Phi_\eta$ are the fixed original positions. They are internal by hypothesis. Apply Theorem 1.8.9 to obtain (1.43) and the two contraction bounds. Compose that identity with $Q_{\zeta, \mathcal{H}_\ell}^{\text{free}}$ and use (1.42) to obtain (1.44). The factors were chosen before ζ . $\square$

1.9.4 Operations beside untouched registers

Definition 1.9.4.1 (Extension by spectator registers). [Lean](#) [✓ Stmt](#) *Given an actual composition $W : a \rightarrow b$ and a spectator list ℓ , move ℓ before a , perform W while leaving those registers unchanged, and move them back after b . Denote the resulting composition $a\ell \rightarrow b\ell$ by W_ℓ .*

Theorem 1.9.4.2 (Exact spectator extension). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The spectator extension has exactly the same ordered source list as W , is allowed whenever W is allowed, and satisfies*

$$W_\ell C_{a, \ell}^{-1} = C_{b, \ell}^{-1} (W \otimes I_{\mathcal{H}_\ell}). \quad (1.45)$$

The operator identity holds without norm or normalization assumptions.

Proof. [✓ Proof](#) On an elementary input $x \otimes y$, the first exchange gives $y \otimes x$, the framed composition gives $y \otimes Wx$, and the last exchange gives $Wx \otimes y$. Linearity gives (1.45). Exchanges create no sources, and leaving a register unchanged preserves the source list and allowedness of the operations on the other registers. $\square$

1.10 Chronological partial source expansions

We follow the chronological expansion in [1, proof of Theorem 5.2], 04-compression.tex, lines 342–434. Each gate has its own participating parties and the common source data constructed above. The following statements establish the exact partial expansion and its scalar cost. The estimate for corrected source operators requires the subsequent separation and second-moment arguments.

Definition 1.10.1 (A chronological composition). Lean Lean Lean ✓ Stmt Let P be any set of parties. A chronological composition from register list a to register list b is formed from identities, consecutive composition, private operations, exchanges of adjacent registers, and nonprivate gates. A gate has a finite participant set C , an injection $C \hookrightarrow P$, and common finite source data on C . It acts on its prescribed input registers and as the identity on all remaining registers. Any composition may also be extended by an idle register. A composition is allowed when each private operation is a contraction. Its complete operator $W : \mathcal{H}_a \rightarrow \mathcal{H}_b$ uses the aggregate operator of each nonprivate gate.

Theorem 1.10.2 (Norm of the complete composition). Lean ✓ Stmt Every allowed chronological composition satisfies $\|W\| \leq 1$.

Proof. ✓ Proof Each gate is a contraction. Consecutive composition, extension by an identity, and unitary identifications of the register order preserve this bound. Induction on the composition proves the claim. □

Definition 1.10.3 (Gate occurrences and source positions). Lean Lean Lean Lean Lean ✓ Stmt Let $\mathcal{G}$ be the finite set of occurrences of nonprivate gates. Repeated uses of the same gate give distinct elements of $\mathcal{G}$. For $g \in \mathcal{G}$, let $C_g \subseteq P$ be its participant set, E_g its ordered source slots, Ξ_g its original branch set, and $c_{g,\xi}$ its original coefficients. The source positions are $\mathcal{E} = \{(g, e) : g \in \mathcal{G}, e \in E_g\}$. Each position has the two original endpoints of its slot, transported along the participant injection. No monomial label enters the definition of $\mathcal{E}$.

Theorem 1.10.4 (Incidence of the original source positions). Lean Lean ✓ Stmt The two endpoints of (g, e) are distinct and both belong to C_g .

Proof. ✓ Proof Each original slot has distinct endpoints in its gate’s participant set. The participant injection preserves distinctness. Consecutive composition retains the occurrence of the gate containing the slot. □

Definition 1.10.5 (The partial branch choices). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Fix affected parties $\mathcal{A} \subseteq P$. A gate occurrence is touched when $C_g \cap \mathcal{A} \neq \emptyset$. Choose one label $\xi_g \in \Xi_g$ at each touched occurrence and no label at the other occurrences. Let $\Xi_{\mathcal{A}}$ be this finite set of choices. In the resulting composition W_{ξ} , touched gates are replaced by their chosen prepared branches, while untouched gates retain their complete operators. Identify all exterior owners as in Definition 1.9.2.1. The coefficient of the partial branch is $c_{\xi} = \prod_{g: C_g \cap \mathcal{A} \neq \emptyset} c_{g, \xi_g}$, and its weighted sum is $\sum_{\xi \in \Xi_{\mathcal{A}}} c_{\xi} W_{\xi}$. An untouched gate has one partial choice with coefficient one, even if its original branch set is empty.

Theorem 1.10.6 (Allowed partial branches and their actual sources). Lean Lean Lean Lean ✓ Stmt If the original composition is allowed, each partial branch is allowed. Its source list is obtained from the chosen sources of the touched gates by retaining exactly those with at least one endpoint in $\mathcal{A}$, and then grouping the exterior owners. Each retained source has its original endpoint spaces and vector. The order and multiplicity of the source occurrences are preserved.

Proof. ✓ Proof A chosen prepared branch is allowed. An untouched gate is a single contraction on exterior memory. The owner identification turns a pair with two exterior endpoints into a local normalized preparation. Induction on the chronological composition gives the asserted source list and allowedness. $\square$

Theorem 1.10.7 (Exact chronological operator identity). Lean Lean Lean Lean Lean
✓ Stmt Let I_a, I_b be the canonical isometries identifying the input and output memories with their grouped-owner memories. Then

$$\left(\sum_{\xi \in \Xi_{\mathcal{A}}} c_{\xi} W_{\xi} \right) I_a = I_b W. \quad (1.46)$$

The private operations need not be contractions for this identity.

Proof. ✓ Proof At a touched gate, sum its original branch expansion and extend the sum by the spectator identity. At an untouched gate, the single partial branch is its aggregate operator. The resulting identity is compatible with consecutive composition, since the two independent finite sums distribute and their coefficients multiply. It is also compatible with adjoining idle registers. Induction gives (1.46). $\square$

Theorem 1.10.8 (Local contributions to the scalar cost). Lean Lean Lean Lean Lean Lean
Lean ✓ Stmt A touched gate contributes $\sum_{\xi \in \Xi_g} |c_{g,\xi}|$ to the sum of absolute partial coefficients. An untouched gate, identity, private operation, or register exchange contributes one. Consecutive compositions multiply these sums. Adjoining an idle register leaves the sum unchanged.

Proof. ✓ Proof Use the defining choice set at each elementary operation. For consecutive compositions, apply $|cd| = |c| |d|$ and distribute the sum over the Cartesian product of the two choice sets. $\square$

Theorem 1.10.9 (The exact product of the gate costs). Lean Lean Lean Lean ✓ Stmt Let $\mathcal{G}_{\mathcal{A}}$ denote the touched gate occurrences and put $t = |\mathcal{G}_{\mathcal{A}}|$. Then

$$\sum_{\xi \in \Xi_{\mathcal{A}}} |c_{\xi}| = \prod_{g \in \mathcal{G}_{\mathcal{A}}} \sum_{\eta \in \Xi_g} |c_{g,\eta}|, \quad (1.47)$$

and the sum of $|c_{\xi} \overline{c_{\zeta}}|$ over all independent ket and bra choices is the square of this product. If each touched gate's coefficient sum is at most B , these two costs are bounded by B^t and B^{2t} , respectively. Every occurrence contributes separately, including repeated uses of one gate.

Proof. ✓ Proof Induction on the chronological composition gives the product formula from the local contributions. Bound each nonnegative factor by B . The ket and bra sums are independent, and complex conjugation preserves absolute value, so their total cost is the square of the first sum. $\square$

Theorem 1.10.10 (The density expansion in orthonormal coordinates). Lean Lean Lean
✓ Stmt Choose finite orthonormal bases of the grouped input and output memories. Let M be the matrix of $I_b W I_a^{-1}$ and K_{ξ} the matrix of the partial branch W_{ξ} . For every input matrix ρ ,

$$M = \sum_{\xi \in \Xi_{\mathcal{A}}} c_{\xi} K_{\xi}, \quad (1.48)$$

$$M \rho M^{\dagger} = \sum_{\xi, \zeta \in \Xi_{\mathcal{A}}} c_{\xi} \overline{c_{\zeta}} K_{\xi} \rho K_{\zeta}^{\dagger}. \quad (1.49)$$

In each summand, the untouched gates retain their complete operators. The matrix ρ need not be positive or normalized.

Proof. ✓ Proof Compose (1.46) with I_a^{-1} and take matrices. Distribute the finite sums in $M\rho M^\dagger$. Taking the adjoint conjugates each bra coefficient. $\square$

Theorem 1.10.11 (Fixed positions for every partial source inventory). Lean Lean Lean Lean Lean ✓ Stmt The positions $\mathcal{E}$ admit a fixed exhaustive ordered list without repetitions, in the preparation order of the actual composition. Consecutive compositions list the later composition's positions first; the original slot order within a gate is retained. Each position e has fixed halfspaces $\mathbb{C}^{u_e}$ and $\mathbb{C}^{v_e}$.

For every partial choice ξ , its source inventory is obtained from this fixed list by retaining exactly those positions with an endpoint in $\mathcal{A}$. A retained position has the grouped original endpoints, the fixed halfspaces, and its normalized chosen branch vector. Consequently the ordered layout of all surviving source registers is independent of ξ .

Proof. ✓ Proof Induct on the chronological composition. The two occurrence sets of consecutive compositions are disjoint, so concatenating their lists preserves uniqueness. At a gate, list its prescribed slots. The participant injection preserves their distinct endpoints, and grouping owners removes a slot exactly when both endpoints are exterior. The remaining fields are its prescribed spaces and vector. Removing vectors from these records leaves a layout determined solely by the original positions and $\mathcal{A}$. $\square$

1.11 Fixed sources for the chronological partial expansion

We now place the sources of every partial branch in the same ordered register spaces. The order is inherited from the original gate occurrences, before any branch labels are chosen. This supplies the common free inputs used in the two separated contractions in [1, proof of Theorem 5.2], 04-compression.tex, lines 342–434.

Definition 1.11.1 (The retained source positions). Lean Lean Lean ✓ Stmt Let $\mathcal{E}$ be the original ordered source positions and let $\mathcal{A}$ be the affected parties. Retain the ordered sublist

$$\mathcal{E}_{\mathcal{A}} = \{e \in \mathcal{E} : p_e \in \mathcal{A} \text{ or } q_e \in \mathcal{A}\}.$$

Write $r = |\mathcal{E}_{\mathcal{A}}|$. Reading this sublist in order gives a bijection from $\{0, \dots, r-1\}$ to its positions. The reference source list assigns to the corresponding position e its grouped endpoints, its fixed halfspaces $\mathbb{C}^{u_e}$ and $\mathbb{C}^{v_e}$, and the zero vector. Thus the reference list specifies common register spaces. Its vectors will be replaced by the normalized vectors of each partial branch.

Theorem 1.11.2 (The common ordered register spaces). Lean Lean Lean Lean Lean ✓ Stmt The source at reference index i has the grouped original endpoints of its enumerated position e_i and the halfspaces $\mathbb{C}^{u_{e_i}}$, $\mathbb{C}^{v_{e_i}}$. The endpoints remain distinct. Every partial branch has precisely this ordered source-register layout, independently of its labels. Membership in the reference list is equivalent to having an affected endpoint.

Proof. ✓ Proof Filter the fixed exhaustive list of original positions by endpoint incidence. Its lack of repetitions gives the stated bijection. At least one endpoint of every retained position is affected, so grouping exterior owners cannot identify its two endpoints. The ordered inventory formula of Theorem 1.10.11 gives the same endpoint spaces for every branch. $\square$

Theorem 1.11.3 (Preparation in the fixed chronological sources). [Lean](#) [✓ Stmt](#) *Let the chronological composition be allowed. For every partial choice ξ there are unit vectors*

$$\eta_{\xi,i} \in \mathbb{C}^{u_{e_i}} \otimes \mathbb{C}^{v_{e_i}}, \quad 0 \leq i < r,$$

and an allowed composition V_ξ containing no pair sources, such that the actual ordered source list of the partial branch consists exactly of these vectors at the reference positions. If P_{η_ξ} prepares them and leaves the original input registers unchanged, then

$$W_\xi = V_\xi P_{\eta_\xi}.$$

The input register list of V_ξ is the common source layout followed by the grouped original inputs. Its output register list is the grouped original output list. Both lists are independent of ξ . The assertion includes an empty source list and an empty set of partial choices.

Proof. [✓ Proof](#) Equal ordered layouts identify the two halfspaces at each source position. Transport the actual normalized source vectors along these identifications. The resulting reference list is exactly the branch's source inventory. The source-preparation theorem moves this inventory to the initial registers and gives an allowed remaining composition containing no sources. Its exact recovery identity is the displayed equality. $\square$

Definition 1.11.4 (The original label and vector at a position). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For a touched gate occurrence g , a partial choice ξ determines its original label $\xi_g \in \Xi_g$. Write $\nu_{e,\alpha}$ for the original vector of source position $e = (g, j)$ at local label $\alpha \in \Xi_g$. At the retained reference index i , with original position $e_i = (g_i, j_i)$, define

$$\eta_{\xi,i}^{\text{orig}} := \nu_{e_i, \xi_{g_i}}.$$

Both sides belong to the fixed coordinate space $\mathbb{C}^{u_{e_i}} \otimes \mathbb{C}^{v_{e_i}}$.

Theorem 1.11.5 (Recovery of the locally selected source vectors). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Every retained source belongs to a touched gate. All vectors $\nu_{e,\alpha}$ have norm one. The actual source record at a retained position is its original endpoint pair, grouped by affected ownership, together with $\eta_{\xi,i}^{\text{orig}}$ in the fixed halfspaces.*

Consequently any exact representation of the ordered source inventory in the reference positions uses precisely these vectors. In Theorem 1.11.3 one may take $\eta_{\xi,i} = \eta_{\xi,i}^{\text{orig}}$. In particular, this vector depends only on the local label at its original owning gate, independently of every other gate label.

Proof. [✓ Proof](#) A retained endpoint belongs to its owning gate, so that gate is touched. Induction on the chronological composition recovers its chosen local label and original vector. At a gate the vector is the given normalized source vector; composition and idle registers preserve its occurrence. Compare the two ordered inventory lists at reference index i . Equality of their source records gives equality of the vectors in their common halfspaces. Substitution in the fixed preparation gives the final assertion. $\square$

Theorem 1.11.6 (Two contractions for the actual sources). [Lean](#) [✓ Stmt](#) *Let W be an allowed composition and divide its parties into two sides. Choose a set of free positions in its actual ordered source list, containing every source that crosses the division. Let P_{fix} prepare the original source vectors at the remaining positions. There are an allowed composition V containing no pair sources, and two allowed compositions A, B containing no pair sources, such that*

$$\|A\| \leq 1, \quad \|B\| \leq 1, \quad J_{\text{out}} V P_{\text{fix}} = (A \otimes B) J_{\text{free}}.$$

Here J_{free} and J_{out} group the input and output registers by side. The equality holds on the entire free input space. The maps V, A, B can be chosen before any vectors are supplied at the free source positions.

More precisely, for every family ζ of vectors at those positions, let P_{ζ}^{free} prepare that family and let $P_{\text{fill}(\zeta)}$ prepare the whole source list, retaining the original vectors at fixed positions. Then

$$J_{\text{out}} V P_{\text{fill}(\zeta)} = (A \otimes B) J_{\text{free}} P_{\zeta}^{\text{free}}.$$

Taking the original source vectors gives the same equality with $V P_{\text{fill}(\zeta)} = W$.

Proof. ✓ Proof Enumerate the actual source list in its original halfspaces. The source-preparation theorem gives $W = V P_{\eta}$ with V allowed and containing no sources. Every fixed source is internal to one side and its original vector has norm one. Apply selective factorization to V and these fixed preparations. Its equality holds before evaluating any free source vectors, and hence specializes both to arbitrary ζ and to the original vectors. $\square$

Bibliography

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